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Jinger Pan, Yue Xi, Dixiao Tan, Yaqian Borogjoon Bao, Victor Kuperman & Ming Yan

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**Running head: TEXT DIRECTION EFFECTS IN CHINESE PASSAGE READING**

**The Corpus of Bidirectional Reading of Traditional Chinese Text: An Extension of the  
Multilingual Eye-Movement Corpus**

Jinger Pan <sup>(1)</sup>, Yue Xi <sup>(1)</sup>, Dixiao Tan <sup>(2)</sup>, Yaqian (Borogjoon) Bao <sup>(3)</sup>, Victor Kuperman <sup>(3)</sup>, and  
Ming Yan <sup>(2, 4)</sup>

(1) Department of Psychology, The Education University of Hong Kong, Hong Kong

(2) Department of Psychology, University of Macau, Macau

(3) Department of Linguistics and Languages, McMaster University

(4) Center for Cognitive and Brain Sciences, University of Macau, Macau

**Author Note**

Correspondence should be sent to Ming Yan (email: mingyan@um.edu.mo).

### **Abstract**

We present a corpus of eye-movement data collected during the reading of passages in traditional Chinese and developed in the framework of the Multilingual Eye Movement Corpus (MECO) project. A distinctive feature of traditional Chinese writing is that texts in this written language can be formatted and read either horizontally or vertically. Leveraging this characteristic bidirectionality, our corpus, based on 60 Hong Kong participants' eye movement data, is the first to offer a within-subject and within-item comparison of eye movements during horizontal and vertical text reading. In addition to the corpus itself, we provide reliability estimates of the eye-tracking data and linear mixed-effects model analyses for its technical validation. The results demonstrate that horizontal and vertical reading of traditional Chinese passages yield comparable spatial and temporal eye movement patterns, contributing to our understanding of both universal and language-specific cognitive, visual, and oculomotor processes in reading. Ultimately, this corpus supports the development of more comprehensive and ecologically valid models of eye movement control in reading.

*Keywords:* eye movement, corpus analysis, traditional Chinese

### Background & Summary

As a complex dynamic system influenced by visual, lexical, and syntactic features of words, eye movement control during reading involves two key decisions: a spatial decision concerning where to direct the eye gaze and a temporal decision concerning when to move the eyes (e.g.,<sup>43</sup>). The former is reflected typically in eye fixation location, whereas the latter is measured by fixation duration. Existing studies have provided extensive insights into eye movements during reading, most of which are focused on alphabetic languages that strictly adopt a horizontally rightward direction of reading and writing<sup>1</sup>. One fascinating factor that affects both learning to read and the everyday practice of reading comprehension is the vast natural diversity in languages.

How does the language environment shape the way we read? The ultimate goal of reading research is to develop theories that explain both the universal and language-specific aspects of reading. Due to its logographic nature and fundamental differences from alphabetic scripts, the Chinese writing system presents unique theoretical challenges for understanding universal and language-specific influences of eye movement control. The basic writing units, Chinese characters, occupy the same squared space irrespective of their visual complexity. The Chinese sentences are a string of continuously written characters without inter-word spaces. Thanks to these features, a number of key findings, which deviate fundamentally from the classical patterns observed in English and other alphabetic scripts, have been reported in Chinese. For example, the perceptual span—the area from which a reader can obtain useful information within a single fixation—is larger in English, covering 14-15 upcoming letters<sup>34</sup>. In contrast, Chinese readers demonstrate a smaller perceptual span, typically encompassing up to four upcoming characters, likely due to the higher text density in Chinese (e.g.,<sup>14, 61</sup>). Another distinctive feature of the

Chinese orthography is its direct representation of meaning, allowing for rapid semantic access without phonological mediation<sup>11, 54</sup>. This characteristic enables early and direct access to semantic features from parafoveally presented words before these words are being fixated upon (e.g.,<sup>50, 60</sup>). In contrast, such effects have been generally elusive in English, where phonological activation precedes semantic processing (e.g.,<sup>13, 15, 44</sup>).

**The corpus-analytic approach of eye movements in reading.** In psycholinguistic research, scholars have developed several methodological approaches, including the experimental control approach, the corpus-analytic approach, and the computational modeling approach, with each offering distinct advantages<sup>21, 23</sup>. The experimental control approach is the most traditional and widely used method, which involves orthogonal manipulation of a small number of factors, typically no more than three. This approach allows researchers to isolate the effects of particular linguistic features. For example, studies have shown that traditional Chinese readers can obtain visual, phonological, and morpho-semantic knowledge from the parafoveal preview<sup>37, 51, 63, 64</sup>). One prominent advantage of the experimental control approach lies in its ability to establish clear cause-and-effect relationships. However, it may sacrifice ecological validity due to its limited number of factors being manipulated simultaneously.

The corpus-analytic approach, in contrast, emphasizes statistical control of naturalistic data using advanced statistical techniques, including linear mixed-effects models (LMMs), to account for variabilities across participants and items. This approach is particularly valuable when working with large datasets, as it allows researchers to generalize findings beyond the confines of tightly controlled experiments. This allows corpus analyses to cover much wider and more complete ranges of independent variables than orthogonally designed experiments. In this case, researchers benefit from the corpus-analytic approach in observing more reliable and

comprehensive trends of the independent variables. Nevertheless, results based on corpus analyses may demand a high level of statistical expertise and careful interpretation, as model predictors can be correlated substantially. It is also important to keep in mind that results based on the experimental control and corpus-analytic approaches can be supportive of each other. For instance, following a long-standing theoretical debate between parallel and serial processing models of reading, Yan et al.<sup>62</sup> reported a morphological complexity effect on first-fixation location in Uighur, an agglutinative Turkic language that is written in a modified Arabic script. In Experiment 1, they conducted a corpus analysis and found an effect of the number of morphological suffixes on first-fixation location, indicating that the readers target closer to the word beginning as morphological complexity increases. In Experiment 2, they orthogonally manipulated the morphological complexity of target words (double-suffixed complex vs. monomorphemic simplex words) that were matched closely on word length, part of speech, word frequency, initial bigram frequency, and predictability. Their Experiment 2 also demonstrated that first-fixation location was closer to the word beginning among the complex words than the simplex words, jointly supporting the notion that morphological boundary information can be obtained parafoveally.

Finally, empirical evidence from both experimental and corpus studies can provide valuable references for computational modeling, which involves constructing formal models that simulate the cognitive, visuo-perceptual, and physiological processes underlying reading. Computational modeling is especially powerful for testing theoretical assumptions and for generating predictions. Together, these three approaches form a robust framework for researchers to develop a comprehensive understanding of the cognitive mechanisms underlying language processing.

Today, eye-tracking corpora during sentence reading are broadly available for many scripts,

including Chinese (simplified<sup>30, 39, 56</sup>; traditional<sup>59</sup>), Dutch<sup>6</sup>, English<sup>45</sup>, French<sup>20</sup>, German<sup>23</sup>, Hindi<sup>12</sup>, Russian<sup>29</sup>, Turkish<sup>36</sup>, Uighur<sup>62</sup>, etc. These data are valuable sources for cross-linguistic comparisons. Generally, across language words that occur more frequently, are shorter in length, and are easier to predict from the preceding context tend to receive shorter fixations (e.g.,<sup>22, 27</sup>). Moreover, the Potsdam Sentence Corpus<sup>21, 22, 23</sup> and its successor, the Beijing Sentence Corpus<sup>39, 56, 59</sup>, incorporated characteristics of the preceding word (N-1) and the following word (N+1) into their models, revealing that the processing of the word that is currently being fixated on can be simultaneously affected by the properties of surrounding words.

Eye-tracking corpora have also been developed for passage reading such as the Ghent Eye-tracking Corpus (GECO; e.g.,<sup>6, 7, 49</sup>). In contrast to sentence reading, passage reading involves longer, more naturalistic texts that require readers to integrate information across multiple sentences and paragraphs. This introduces additional cognitive demands such as discourse-level processing and memory integration. As a result, eye movements during passage reading tend to be more variable, leading to overall longer fixations and more regressions (e.g.,<sup>42</sup>). Other eye-movement corpora of passage reading include CELER<sup>4, 18, 31</sup>.

Another relevant development in the corpus collection is the availability of multi-lingual corpora that enable direct cross-linguistic comparisons, also between different types of written languages and writing directions. For example, the Multilingual Eye-Movement Corpus (MECO) is the data resource that contains eye-tracking data on first-language passage reading from native readers of 23 unique languages across its Wave 1 and Wave 2 releases<sup>2, 46, 47</sup>, as well as English as a second language passage reading data from the same participants<sup>25, 26</sup>. One of its key findings is that the average fixation duration is relatively stable across languages, while metrics like skipping vary significantly and systematically, correlating with the average word length in

the written language. In summary, passage reading offers a more ecologically valid view of how people read in real-world contexts.

**The influence of text direction on eye movements.** Studies reviewed above almost exclusively focused on horizontal reading, due to the widespread use of the horizontal text layout across languages. One of the rare exceptions is traditional Mongolian, where texts are written and read vertically (e.g.,<sup>2, 5, 48, 57</sup>). Recently, another critical feature of the Chinese language that has attracted significant attention among psycholinguistic researchers involves its flexibility in text direction. Traditionally, Chinese was written for centuries in vertical columns from top to bottom, with each new column of text beginning to the left of the previous one. One possible reason for this vertical text layout is that bamboo and wooden strips were the main media for writing when the first standardized character set, the Small Seal Script, was adopted as the official script of China in the Qin dynasty (221 – 206 B.C). In fact, the horizontal text layout commonly seen today in Chinese writing was only adopted about a century ago, initially to accommodate scientific formulas within the text. In places where traditional Chinese is used, vertical text is still frequently found in novels, newspapers, and magazines, whereas horizontal alignment is more typical in science books and textbooks for other subjects. As a result, traditional Chinese readers are generally familiar with both vertical and horizontal formats, although individual readers may differ in their experience with vertical reading<sup>55</sup>.

Pan et al.<sup>40</sup> examined processing of parafoveal information during the vertical reading of traditional Chinese sentences and reported preview effects that were comparable to previous horizontal reading studies, indicating that lexical access can be fundamentally similar in horizontal and vertical reading. Due to the flexibility in text direction, some other studies adopted a within-subject and within-item comparison of text direction, in order to further



distinguish between the effects of general language proficiency and direction-specific reading experience. For instance, Yan et al.<sup>58</sup> compared reading performance between horizontal and vertical reading among Taiwanese readers who were proficient in both layouts. They found that reading speeds and eye movement parameters were largely comparable regardless of direction. This finding aligns with earlier research on Japanese reading (e.g.,<sup>35</sup>). Importantly, Yan et al.<sup>58</sup> identified a trade-off between fixation duration and saccade amplitude between horizontal and vertical reading. Specifically, vertical reading was associated with longer fixations and longer saccades, suggesting that readers familiar with vertical text may process more information per fixation during vertical reading. In a subsequent study, Yan et al.<sup>55</sup> examined the horizontal and vertical perceptual span among readers of traditional Chinese from Hong Kong. Their results showed that vertical reading was associated with an overall smaller perceptual span, in line with Borjigin et al.<sup>5</sup>, likely due to participants' relatively lower familiarity with vertical text. However, those participants with greater vertical reading experience demonstrated an expanded perceptual span exceeding four upcoming characters. The results suggest that oculomotor activities can improve with direction-specific perceptual experience, challenging the assumption that vertical reading is inherently less efficient. A follow-up study by Hao and Yan<sup>9</sup> demonstrated that simplified Chinese readers, who were accustomed only to horizontal reading, showed less effective eye movements during their vertical than horizontal reading of second-language Japanese sentences. Meanwhile, their vertical eye movements developed more rapidly as a function of L2 fluency than horizontal eye movements. As such, direction-specific perceptual experience from the first language can transfer across different writing systems to influence second language acquisition.

**The present study.** Combining the research ideas reviewed above, here we report the

Bidirectional Chinese MECO corpus, as the first eye tracking corpus of passage reading that involves a within-subject and within-item comparison of text direction. Data were recorded from 60 Hong Kong participants when reading 12 passages in traditional Chinese.

## Method

### Participants

Sixty students from the Education University of Hong Kong ( $M_{age} = 20.9$  years,  $SD = 2.7$ , 48 females) with normal or corrected-to-normal vision participated in the eye-tracking experiment. They were recruited through the university's mailing system and poster advertisements. To ensure the language dominance of traditional Chinese, all of the participants had undertaken their education in Hong Kong since primary school. Experimental procedures were approved by the Human Research Ethics Committee of The Education University of Hong Kong (Ref. no. A2022-2023-0515-01). All methods were carried out in accordance with relevant guidelines and regulations. Participants signed a written informed consent on data collection and publication of unidentifiable data prior to the experiment in the eye-tracking lab. The data collected were saved using anonymous codes, which were not linked to the identifiable information of participants. Participants' information was obtained only for data collection scheduling purposes and was kept in a password-protected computer. The sample size was identical to previous studies on eye movement corpora (e.g.,<sup>39, 59</sup>) and was supported by recent simulation work by Kumle et al.<sup>24</sup>. Using a 5-point Likert scale, the participants were asked to rate how familiar they were with vertical reading (1 = very unfamiliar, 5 = very familiar). The average rating for familiarity with vertical reading was 3.3 ( $SD = .84$ ), and was comparable to the sample reported in Yan et al.<sup>55</sup>.

## Material

The participants read 12 experimental passages written in traditional Chinese. These texts followed the format and content of the texts in the MECO project that the readers read in their first language, MECO-L1<sup>46</sup>. They resembled Wikipedia-style encyclopedia entries and covered a wide range of topics, such as historical figures, events, and natural or social phenomena. The topics were selected to avoid requiring specialized academic knowledge and to minimize cultural bias, ensuring equal familiarity across different study sites. Originally, the texts were composed in English, loosely inspired by Wikipedia articles. Five of the 12 texts were selected for translation into Chinese by the MECO research team in Taiwan<sup>47</sup> and back-translation was used to ensure translation accuracy and content consistency. The other seven texts were not translated, but related to the corresponding English passages in that they were on similar topics, written in the same expository style, and matched in length and difficulty. Since the traditional Chinese texts were initially tailored for a Taiwanese audience, minor vocabulary adjustments were made to better suit participants from Hong Kong. The present sample of participants only read the passages in their native language and script (i.e., traditional Chinese) and did not read the English texts.

The final Bidirectional Chinese MECO materials contained 12 passages, with 4 - 8 sentences ( $M = 5.83$ ,  $SD = 1.27$ ), 144 to 194 words ( $M = 174.58$ ,  $SD = 16.22$ ), and 232 to 338 characters ( $M = 293.42$ ,  $SD = 30.89$ ) per passage. We have followed lexical entries in Chinese dictionaries to determine word boundaries<sup>17</sup>. In terms of word length distribution, among the 2095 word tokens, there were 808 one-character words, 1171 two-character words, and 116 words that were three or more characters long. In terms of visual complexity, the number of strokes in a word (which may contain multiple characters) varied from 1 to 58 ( $M = 15.74$ ,  $SD =$

8.66). The numbers of words with strokes in the ranges of 1–10, 11–20, 21–30, and 31 or above were 757, 700, 530, and 108, respectively. Word frequency, based on statistics of a modern Chinese corpus<sup>17</sup>, varied from 0.08 to 24098.25 occurrences per million ( $M = 2300.38$ ,  $SD = 6264.43$ ). The numbers of words with frequencies in the ranges of below 50, 51–200, 201–500, 501–1000, and 1001 or above were 805, 329, 211, 245, and 505, respectively. The numbers of words with log-transformed frequencies in the ranges of below 1, 1–2, 2–3, and 3 or above were 460, 528, 602, and 505, respectively.

Additional materials, administered across all samples in the MECO project, included the nonverbal IQ test from the Culture Fair Test-3 (CFT20, Subset 3 Matrices, short version, Form A, timed at 3 minutes<sup>52</sup>), and an abridged version of the Language Experience and Proficiency Questionnaire (LEAP-Q<sup>33</sup>). The CFT20 aimed at providing a comparable measure of nonverbal intelligence, and the LEAP-Q at collecting basic demographic and linguistic information about participants<sup>46</sup>.

### Apparatus

The participants' eye movements were recorded with an EyeLink 1000 Desktop system running at a sampling rate of 1,000 Hz. The passages were presented on a 24-inch ASUS VG248QE monitor (resolution: 1920-1080 pixels; frame rate: 144 Hz) using the PMingLiU font (monospaced). The participants were seated comfortably with their heads placed on a chin-and-forehead rest at a distance of 70 cm from the monitor. Each character occupied a 43-by-43 pixel grid and subtended approximately 1.0 degrees of visual angle. The distance between the tops of every two consecutive lines was set to 80 pixels (i.e., approximately double spacing). This wide inter-line spacing was used to compensate for possible small calibration errors. With the current setup, we aimed to achieve precise alignment of fixations with the correct line of text. While

wider spacing may further reduce fixation misalignment, it can also disturb readers' typical reading patterns. All recordings and calibrations were performed monocularly based on the participants' right eyes, and viewing was binocular.

## Procedure

The participants were instructed to read the passages silently for comprehension and then press a keyboard button to signal completion. The experiment began with one practice trial, which was not included in the analyses reported below. Each participant received the 12 experimental passages in a different, randomized order. Of these, six passages were read horizontally and six vertically, with the assignment of reading orientation randomized across passages. The presentation of passages in the same reading orientation was blocked. The order of the reading orientation was counterbalanced across participants. The participants' gaze positions were calibrated using a standard 9-point grid (error  $< 0.5^\circ$ ) at the beginning of each block. A drift check with a fixation point occupying the same position as the first character of the passage was conducted before a new passage was presented. An extra calibration was carried out when participants failed to pass the drift check. To reduce calibration and drift errors, the participants were instructed to maintain a stable posture, which was monitored by the experimenter. Each passage was presented across two consecutive screens, each containing a maximum of 11 lines of text. While this setting prevented the readers from regressing to the beginning of the passage, we retained the large line spacing to ensure data quality. To ensure that the horizontal and vertical layouts were comparable, each line of text was identical across the two conditions (see Figure 1). Each passage was followed by four easy yes-no comprehension questions to encourage the participants' engagement with the reading task. The comprehension questions were included to encourage the participants to read carefully, as reading behavior may vary

depending on the comprehension demands. The average percent correct across participants was 88.6% ( $SD = 5.5\%$ ).

— FIGURE 1—

## Data Processing

Our analyses below are based on a subset of data that underwent several trimming procedures. As part of trimming, six trials containing eye-tracker errors were excluded from the analyses. Additionally, the first and the last fixated words in each line of text were discarded, because fixations on these words do not reflect typical lexical processing and often are outcomes of the return-sweeps from previous lines. Words with first-fixation durations (duration of the first fixation on a word, irrespective of total number of fixations) shorter than 60 ms or longer than 800 ms and gaze durations (the sum of all fixation durations during the first-pass reading of a word) longer than 1500 ms were removed. Together the analytical dataset consisted of 31937 observations.

## Data Records

The Bidirectional Chinese MECO data is publicly available from the Open Science Framework (OSF) repository under the CC BY 4.0 license<sup>38</sup>. The dataset includes eye-tracking and reading comprehension data. We acknowledge that data selection criteria may differ among researchers. Therefore, we have retained all observations in the data file, allowing readers to apply their own criteria: Our own trimming procedures are described above. We also are aware that different labs make use of different data formats: Our reports are prepared in several formats to satisfy these requirements: All of these formats provide digital files that can be loaded in the R

programming environment.

One reporting format made use of the standard reporting tools available in Data Viewer software<sup>8</sup> (version 4.4.1). The **Data Viewer Reports** folder contains the original reports generated directly by the Data Viewer software without modifications. The files in this folder include: ‘SaccadeReport.rda’, ‘FixationReport.rda’, and ‘IA\_Report.rda’. These files retain the original variables and variable names generated by the Data Viewer software. The variable names can be accessed in the Data Viewer manual. Since Data Viewer software is designed to calculate some spatial oculomotor metrics for horizontal reading only, we recalculated those metrics for both the horizontal and vertical reading based on the X and Y coordinates of fixations and interest area boundaries (extending to the mid-point between lines). Thus, we added variables labeled `firstfix.sac.in` (incoming saccade length in pixels for the first fixation on the IA); `firstfix.sac.out` (outcoming saccade length in pixels for the first fixation on the IA); `firstfix.launch` (launch site distance, incoming saccade length until the space before the IA in pixels); `firstfix.land` (landing position in pixels of the first fixation on the IA) and `firstfix.cland` (landing position of the first fixation relative to the center of the IA). Other additional user-defined variable names of relevance are *direction* (H or V, for horizontal versus vertical), *number* (passage ID from 1 to 12) and *screen* (screen number 1 or 2).

To align the Bidirectional Chinese Corpus data with the MECO project structure and to follow its conventions for variable naming, the interest area reports have also been recalculated and relabeled to match the MECO variables. Therefore, the data storage contains the second folder: **MECO-Aligned Reports**.

MECO-Aligned Reports contain eye-tracking data that have been recalculated and reformatted to match the MECO project’s variable structure. These reports enable a seamless

integration of traditional Chinese data with the data available in MECO for all other languages and makes cross-linguistic comparisons that include traditional Chinese possible. A detailed legend document is included to explain the conversion formulas applied to the original Data Viewer variables.

— TABLE 1—

The MECO-Aligned Reports folder is organized into several subfolders. Under the ‘auxiliary files’ folder, the ‘descriptive stats’ subfolder contains the file ‘descriptive\_eyemove.csv’, which provides descriptive analyses of eye movement measures reported at the word level, broken down by direction as presented in Table 1. It also contains the file “correlation\_matrices.xlsx” with Pearson correlations between behavioral measures of reading, calculated from participant-level means for word-level reports. The other subfolder is ‘reading materials,’ which includes the original texts, images used for displaying text and questions, all back-translations in English, text readability and complexity metrics, and similarity scores. The ‘code’ folder includes the R script used for data analysis (Main.R). The ‘primary data’ folder includes ‘comprehension data,’ ‘eye-tracking data,’ and ‘individual differences data’. The individual differences tests include the CFT nonverbal IQ test and the Language Experience and Proficiency Questionnaire, as well as the ratings of familiarity with vertical reading. These data were collected on the same day after the eye-tracking session. The ‘IAS files’ folder includes the interest area files used in Data Viewer for each text at the sentence and word levels. Detailed information is listed below:

- primary data
- comprehension data



- `ch_t_bi_all_acc.rda` (answers for each question by each participant; each participant answers a total of 48 questions, with 4 questions per trial.)
- `ch_t_bi_all_acc_HV.rda` (answers for each question by each participant; broken down by texts read in the horizontal vs vertical direction.)
- `ch_t_bi_br_acc.rda` (accuracy for each trial by each participant)
- `ch_t_bi_acc.rda` (accuracy rate for each participant)
- eye tracking data
  - `eye_data_trimmed_ch_t_bi.rda` (eye movement measures for each word, interest area report at the word level)
  - `ch_t_bi_readrate.rda` (reading rate in words per minute, calculated per each participant and text)
  - `passage_data.csv` (eye movement measures summed by trial)
- individual differences data
  - CFT20 IQ: CFT 20 IQ test responses to each question by each participant, along with total scores for each participant
  - language questionnaire: each participant's responses to the Language Experience and Proficiency Questionnaire
  - vertical reading familiarity: vertical reading familiarity rating

Finally, the root directory of the OSF folder for this corpus contains a folder labeled IAS files:

- `word_IAS` (interest area files for each trial at the word level)

The root directory also contains the `MECO_HK_WordInfo_s.csv` file with lexical characteristics

of all words. The file has the following structure:

- Column 1 (Passage): Passage number (1 to 12)
- Column 2 (Page): Page number (1 to 2)
- Column 3 (WordNr): Word number, the ordinal position of the word within the passage
- Column 4 (SenNr): Sentence number, the ordinal position of the sentence within the passage
- Column 5 (Word): Text of the word
- Column 6 (Length): Word length, number of characters in the word
- Column 7 (WordStroke): Stroke count of the word
- Column 8 (WordFreq): Frequency of the word, number of occurrences per million
- Column 9 (WordID): A unique word ID, encompassing passage, sentence, and word numbers

Furthermore, we provide the R scripts used for LMM analyses reported in the Technical Validation section (“model.R”).

### Technical Validation

We applied and reported two methods of validating the data quality of the Bidirectional Chinese MECO corpus: Reliability estimates of the eye movement measures and statistical regression models aiming to replicate well-established linguistic and visuo-perceptual effects on eye movements during reading.

*Validation of Reliability.* This section follows the procedure developed for the MECO

project in Siegelman et al.<sup>47</sup> and Bao et al.<sup>2</sup>. The reliability was estimated using the split-half technique at both the participant level and the word token level. The within-participants stability of each eye movement measure was estimated by splitting all words of the Chinese texts into “odd” and “even” words, based on their ordinal position in the text. Mean values for a selection of oft-used eye-movement measures were calculated for the even and odd words and their correlation was taken as the estimate of split-half reliability. The only exception to this procedure was the estimation of reliability for reading rate: It was assessed by calculating the intra-class correlation coefficients (ICC) as a metric of agreement in reading rates across the 12 texts. Similarly, to estimate reliability at the level of word tokens, we examined the correlation between means for “odd” and “even” participants within each word token for each eye movement measure. For both participant-level and word token-level reliability, we computed raw correlations and Spearman–Brown-corrected values. As the next step, we calculated within-participants and within-items reliability for horizontal and vertical reading separately. Table 2 summarizes the results for the full data set.

— TABLE 2—

Correlations in eye movement measures at the participant level were extremely high (Spearman-Brown corrected  $r_s \geq 0.96$ ), as shown in Table 2. That is, these measures are extremely stable within participants. The ICC for reading rate was 0.98. At the word token level, the reliability of estimates was lower (Spearman-Brown corrected mean  $r_s = 0.81$ ), especially for first fixation duration and gaze duration. This discrepancy between participant- and word token-level variability replicates findings in other languages represented in MECO.

The estimates for horizontal and vertical readings were estimated separately. All of them

were within  $\pm 0.02$  of the estimates reported in Table 2 for the full sample. Thus, we conclude that the present data are reliable for studying both reading directions.

*Regression models.* To validate the collected data, we report statistical models of temporal eye-movements predicted by major, well-established linguistic and visuo-perceptual factors. In corpus analyses, predictors are often correlated, sometimes substantially. Convergence of the estimated effects for these factors with those reported in the rich existing literature would signal the validity of the present dataset. As a result, zero-order correlation can be affected by uncontrolled confounding factors and potentially obscures genuine effects. To address this issue, LMMs have been proven effective statistical tools, which allow researchers to estimate the influences of correlated predictors, while accounting for variabilities at the levels of participants, words, and sentences. The partial effects from LMMs (i.e., estimates after the statistical control of confounding variables) can offer accurate insights into the relationships of interest in corpus analyses<sup>23</sup>.

We report below LMM estimates using the *lme4* package<sup>3</sup> (Version 1.1-23) and the *lmerTest* package<sup>28</sup> (Version 3.1-2) in the R environment<sup>41</sup> (Version 4.4.1). The fixed-effects included the experimental condition of text direction and linguistic variables of word length, log-transformed word frequency, visual complexity, and all their two-way interactions on gaze duration (i.e., the sum of all fixation durations during the first-pass reading of a word). Generally, launch site – the distance between the location of the last fixation and the beginning of the currently fixated word – cannot be controlled prior to the experiment and has shown robust influences on eye-movement parameters. Therefore, linear and quadratic trends of the launch site were included as model covariates for statistical control. We chose gaze duration as the main dependent variable because it is perhaps the most commonly used index in eye-movement

research and has been shown to be sensitive to visual and linguistic processes in reading<sup>16, 19</sup>. All continuous predictors were centered before entering the model. We included subject- and item-related variance components for the intercepts and random-slopes for the fixed-effects and started with full random-effects. Following a standard parsimonious LMMs selection procedure<sup>32</sup>, we dropped correlation parameters and small variance parameters for successful model convergence. LMM output is reported in Table 3.

— TABLE 3—

Replicating patterns observed during simplified and traditional Chinese reading in previous studies (e.g.,<sup>30, 39, 59</sup>), the current data also showed foveal influences of word frequency, visual complexity, and word length on fixation duration (Table 3 and Figure 2). Overall, fixations were longer for less frequent, longer, and visually more complex words. The word frequency effect is a well-established phenomenon in eye movement research. Readers typically spend less time fixating on high-frequency words, suggesting that these words are recognized more easily. The effect is robust across different languages and writing systems, indicating its fundamental role in lexical processing, and has been implemented in computational models of eye movements, such as the E-Z Reader and SWIFT. The interaction between reading direction and word frequency was not significant, suggesting that lexical access and visual decoding can be fundamentally similar when reading Chinese text in different directions. Similarly, the word length effect is another prominent finding, showing that longer words tend to receive longer fixation durations and to be fixated on more often. This effect is attributable to the increased visual and cognitive processing demands associated with longer words, which often require more time for recognition, as their extended visual span makes them harder to process parafoveally. The word

length effect has also been observed across various orthographies, suggesting it is a general feature of the reading process. Moreover, the word length effect on gaze duration was stronger for vertical reading than for horizontal reading. Finally, the visual complexity effect is rather unique in logographic writing systems like Chinese and Japanese. With an increase in the number of strokes, characters and words tend to elicit longer fixations, reflecting the increased processing demands related to visual decoding. Understanding the visual complexity effect is crucial for developing accurate models of reading in non-alphabetic scripts. The visual complexity effect was also stronger for vertical reading than for horizontal reading.

— FIGURE 2—

To sum up, the present study established an eye movement corpus during the horizontal and vertical reading of traditional Chinese sentences. The core patterns of oculomotor behaviors are fundamentally similar between the two reading directions among adult readers. The sentences of these eye movement corpora, such as MECO, GECO, and Potsdam Sentence Corpus (PSC), and Beijing Sentence Corpus (BSC), have been widely used for different research topics such as the perceptual span and reading development. The current Bidirectional Chinese MECO is well-suited for exploring similar research questions. Although vertical text is not frequently used worldwide, the contribution of Bidirectional Chinese MECO goes far beyond a purely theoretical exploration of text direction effects on oculomotor activities during reading. Many orthographies in Asia have historically adopted vertically written languages, such as Chinese, Japanese (e.g., <sup>9</sup>), Korean, Manchu, and Mongolian, many of which are still actively used. Additionally, older adults are generally more familiar with vertical text. Understanding the cognitive processing across different text directions can be potentially helpful in the development of reading-related

intervention programs for older adults.

#### **Code availability**

The code used in this study is available publicly at the OSF page (<https://osf.io/jyn8s>).

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### **Author Contributions**

Conceptualization: JP, VK, & MY

Methodology: JP, YB, VK, & MY

Validation: VK & MY

Formal Analysis: DT, YB, VK & MY

Investigation: JP & YX

Resources: JP

Writing – Original Draft: MY

Writing – Review & Editing: JP, VK, & MY

Visualization: YX & MY

Supervision: JP

Project administration: JP

Funding acquisition: JP, VK, & MY

### **Competing Interests**

The authors declare no competing interests.

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#### **Ethic Statement**

The study was approved by the Human Research Ethics Committee of The Education University of Hong Kong (Ref. No. 2022-2023-0515).

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Table 1

*Descriptive statistics for eye movements, broken down by direction of reading*

| Variable            | Horizontal |        |       | Vertical |        |       |
|---------------------|------------|--------|-------|----------|--------|-------|
|                     | Mean       | SD     | n     | Mean     | SD     | n     |
| dur                 | 328        | 205    | 26337 | 388      | 235    | 32584 |
| firstfix.cland      | -2.15      | 25.75  | 33633 | -5.33    | 25.27  | 41511 |
| firstfix.dur        | 229        | 101    | 26337 | 257      | 109    | 32584 |
| firstfix.land       | 40.74      | 28.12  | 33633 | 36.80    | 26.45  | 41511 |
| firstfix.launch     | -38.21     | 299.10 | 32984 | -31.23   | 249.55 | 40869 |
| firstfix.sac.in     | 222.92     | 194.81 | 32984 | 165.20   | 181.99 | 40869 |
| firstfix.sac.out    | 193.49     | 185.06 | 33375 | 138.50   | 168.89 | 41322 |
| firstrun.dur        | 259        | 147    | 26337 | 320      | 186    | 32584 |
| firstrun.gopast     | 523        | 1018   | 26337 | 571      | 1316   | 32584 |
| firstrun.gopast.sel | 285        | 169    | 26337 | 348      | 206    | 32584 |
| firstrun.nfix       | 1.14       | 0.40   | 26337 | 1.25     | 0.52   | 32584 |
| firstrun.refix      | 0.12       | 0.33   | 26337 | 0.21     | 0.41   | 32584 |
| firstrun.reg.in     | 0.30       | 0.46   | 26337 | 0.22     | 0.41   | 32584 |
| firstrun.reg.out    | 0.17       | 0.38   | 26337 | 0.14     | 0.35   | 32584 |
| firstrun.skip       | 0.76       | 0.43   | 60791 | 0.68     | 0.47   | 61734 |
| nfix                | 0.62       | 0.86   | 60791 | 0.80     | 0.94   | 61734 |
| nrun                | 0.55       | 0.73   | 60791 | 0.65     | 0.72   | 61734 |
| refix               | 0.14       | 0.35   | 60791 | 0.20     | 0.40   | 61734 |
| reg.in              | 0.30       | 0.46   | 26337 | 0.22     | 0.41   | 32584 |
| reg.out             | 0.31       | 0.46   | 26337 | 0.23     | 0.42   | 32584 |
| reread              | 0.10       | 0.30   | 60791 | 0.11     | 0.31   | 61734 |
| singlefix           | 0.88       | 0.33   | 26337 | 0.79     | 0.41   | 32584 |
| singlefix.dur       | 227        | 99     | 23170 | 257      | 109    | 25776 |
| skip                | 0.57       | 0.50   | 60791 | 0.47     | 0.50   | 61734 |

Note. Means, standard deviations, and sample sizes of various eye-movement variables. The

legend for variable names is available in the OSF repository under MECO-Aligned reports > primary data > eye-tracking data.

refix: Variable indicating whether the IA has been refixated at least once during a trial

reg.in: Variable indicating whether there was at least one regression into the IA

reg.out: Variable indicating whether there was at least one regression from the IA

dur: Total time the interest area was read during the trial in ms (total reading time)

firstrun.skip: Variable indicating whether the IA was skipped during first-pass reading

firstrun.nfix: Number of fixations made on the IA during first-pass reading

firstrun.refix: Variable indicating whether the IA was refixated during first-pass reading

firstrun.reg.in: Variable indicating whether there was a regression into the IA during first-pass reading

firstrun.reg.out: Variable indicating whether there was a regression from the IA during first-pass reading

firstrun.dur: Time the IA was read during first-pass reading (gaze duration)

firstrun.gopast: Sum of all fixations durations from the time the IA was entered until it was left to the right (go-past time/regression path duration)

firstrun.gopast.sel: Sum of all fixations on the IA from the time the IA was entered until it was left to the right (selective go-past time: go-past time minus the time of the regression path)

firstfix.sac.in: Incoming saccade length (in pixels) for the first fixation on the IA. NOTE: In most other MECO files, this metric is given in letters. Exercise caution when merging.

firstfix.sac.out: Outcoming saccade length (in pixels) for the first fixation on the IA. NOTE: In most other MECO files, this metric is given in letters. Exercise caution when merging.

firstfix.launch: Launch site distance (incoming saccade length until the space before the IA) in pixels. NOTE: In most other MECO files, this metric is given in letters. Exercise caution when merging.

firstfix.land: Landing position (pixels) of the first fixation on the IA. NOTE: In most other MECO files, this metric is given in letters. Exercise caution when merging.

firstfix.cland: Centered landing position of the first fixation on the IA (landing position - (Right - Left)) / 2) for horizontal or (landing position - (Bottom - Top)) / 2) for vertical reading. In pixels.

firstfix.dur: Duration of the first fixation on the IA (first fixation duration)

singlefix: Variable indicating whether the IA was fixated only once during first-pass reading



singlefix.sac.in: Incoming saccade length (in letters) for the first fixation on the IA when it was fixated only once during first-pass reading

singlefix.sac.out: Outgoing saccade length (in letters) for the first fixation on the IA when it was fixated only once during first-pass reading

singlefix.launch: Launch site distance (incoming saccade length until the space before the IA) for the first fixation on the IA when it was fixated only once during first-pass reading

singlefix.land: Landing position (letter) of the first fixation on the IA when it was read only once

singlefix.cland: Centered landing position of the first fixation on the IA when it was read only once

singlefix.dur: Duration of the first fixation on the IA when it was read only once

**Table 2***Reliability estimates at the participant level*

| Eye movement measure    | Participant-level reliability uncorrected | Participant-level reliability (Spearman-Brown correction) | Word-level reliability uncorrected | Word-level reliability (Spearman-Brown correction) |
|-------------------------|-------------------------------------------|-----------------------------------------------------------|------------------------------------|----------------------------------------------------|
| Skipping rate*          | 0.97                                      | 0.98                                                      | 0.83                               | 0.91                                               |
| Number of fixations     | 0.98                                      | 0.99                                                      | 0.91                               | 0.95                                               |
| Total fixation time     | 0.98                                      | 0.99                                                      | 0.71                               | 0.83                                               |
| Regression rate         | 0.92                                      | 0.96                                                      | 0.89                               | 0.94                                               |
| First fixation duration | 0.97                                      | 0.99                                                      | 0.28                               | 0.44                                               |
| Gaze duration           | 0.96                                      | 0.98                                                      | 0.48                               | 0.65                                               |
| Rereading rate          | 0.98                                      | 0.99                                                      | 0.86                               | 0.92                                               |

\*This estimate relates to overall skipping rate (whether the word was fixated even once during reading), which is a more robust measure for the reading of extended texts than first-run skipping (see discussion in <sup>46</sup>).

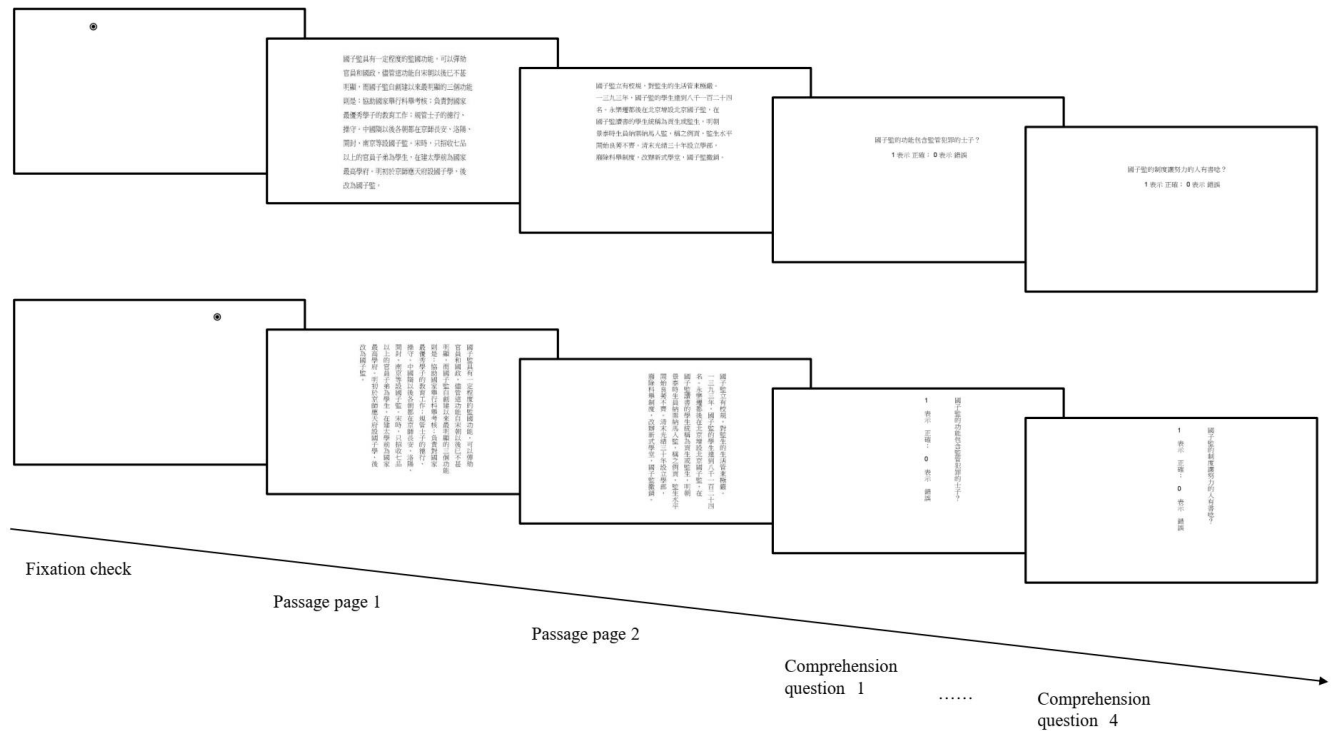
**Table 3***Linear mixed model estimates for gaze duration*

| <b>Fixed effect</b>       | <b>Est.</b> | <b>SE</b> | <b>t-value</b> | <b>p-value</b> |
|---------------------------|-------------|-----------|----------------|----------------|
| (Intercept)               | 5.548       | 0.019     | 295.974        | < 0.001        |
| poly(ls.c, 2)1            | 4.703       | 0.481     | 9.774          | < 0.001        |
| poly(ls.c, 2)2            | -4.250      | 0.430     | -9.882         | < 0.001        |
| len.c                     | 0.075       | 0.007     | 10.188         | < 0.001        |
| lwf.c                     | -0.011      | 0.001     | -8.369         | < 0.001        |
| vc.c                      | 0.056       | 0.007     | 8.617          | < 0.001        |
| hv                        | 0.243       | 0.016     | 15.456         | < 0.001        |
| len.c: hv                 | 0.057       | 0.014     | 4.179          | < 0.001        |
| lwf.c: hv                 | 0.004       | 0.002     | 1.899          | 0.058          |
| vc.c: hv                  | 0.041       | 0.012     | 3.401          | < 0.001        |
| <b>Variance component</b> | <b>SD</b>   |           |                |                |
| WrdID - (Intercept)       | 0.060       |           |                |                |
| SubID - hv                | 0.112       |           |                |                |
| SubID - (Intercept)       | 0.133       |           |                |                |
| ItmID - (Intercept)       | 0.016       |           |                |                |
| Residual                  | 0.417       |           |                |                |

Note. ls.c = launch site, len.c = length of word, lwf.c = log frequency of word, vc.c = stroke count of word, hv = the manipulation of horizontal or vertical reading, WrdID = word-related variance component, SubID = subject-related variance component, and ItmID = passage-related variance component.

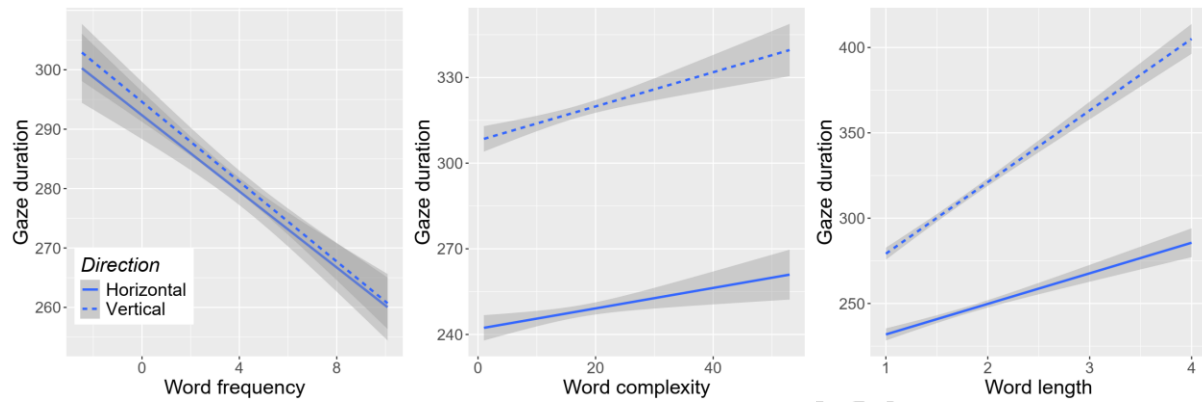
**Figure 1**

*Example trials in horizontal and vertical reading, respectively.*



**Figure 2**

*Gaze duration as a function of text direction and word properties.*



Partial effects of LMMs are plotted using the `remef`<sup>10</sup> (version 0.6.10) and the `ggplot2` packages<sup>53</sup> (version 2.1.0). Shaded bands for partial effects represent 95% CIs based on observation-level LMM residuals.